

Detailed Thesis Structure

Content Guide for a PMSM Control Platform Bachelor Thesis

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Revised structure based on the thesis plan and DHBW guidelines

Purpose

This guide defines the intended content, boundaries and evidence for every proposed thesis section. It is a planning document, not a mandatory writing template.

Editorial Notes

- Corrected terminology: “Permanent-Magnet Synchronous Motors” is the conventional English form.
- Corrected typographical errors: “Commissioning” and “Functional Platform Verification”.
- Corrected Chapter 8 numbering to 8.1-8.5. The original 8.1.1-8.1.5 would require a parent Section 8.1.
- Section 9.3 is retained because it was requested, but it may be redundant when both German and English abstracts and a conclusion are required.
- The proposed wording describes what each section should achieve. Final depth should follow the evidence actually available at submission.

Recommended narrative: Problem -> theory -> existing work -> method -> simulation -> platform design -> experimental evidence -> interpretation -> conclusion.

Primary comparison objective: Use speed control as the common outer-loop objective. Treat torque and position control as secondary unless reliable measurement and validation are available.

Suggested Page Budget

Chapter	Target pages
1 Introduction	3-4
2 Theoretical Background	10-12
3 State of the Art	5-7
4 Methodology	7-9
5 Modelling and Simulation	7-9
6 Control Platform Design	13-16
7 Experimental Verification and Evaluation	13-16
8 Discussion	4-6
9 Conclusion and Summary	2-4
Total main text	64-83; target approximately 65-75

1 Introduction

Chapter purpose: Establish the practical problem, scientific objective and boundaries of the thesis. The chapter should orient the reader without anticipating detailed theory, design decisions or results.

1.1 Context and Motivation

Include:

- Describe the need for a transparent and configurable platform for evaluating permanent-magnet synchronous motors (PMSMs).
- Explain the limitations of closed commercial actuator controllers and black-box motor-control systems: restricted access to internal variables, fixed control algorithms, limited parameterization and poor transferability to unfamiliar motors.
- Introduce the industrial and engineering relevance of rapid motor assessment on common hardware.
- Motivate why six-step commutation, sinusoidal commutation and field-oriented control should be implemented and evaluated on the same platform.
- Conclude with the concrete problem addressed by the thesis rather than a broad history of electric drives.

Writing guidance: Use a short problem-solution figure showing the transition from closed controllers to an open modular development platform.

1.2 Research Objectives

Include:

- State the overall objective: development, commissioning and evaluation of a modular embedded PMSM control platform.
- Separate the engineering objective from the scientific evaluation objective.
- Formulate one primary research question on differences in control performance, current quality, implementation demand and suitability for rapid motor evaluation.
- Add only a small number of supporting questions concerning platform architecture, real-time implementation and transferability.
- Define the expected outputs: custom inverter hardware, sensing, embedded firmware, simulation models, three control methods and comparative measurements.

Writing guidance: Write the objectives as verifiable outcomes. Avoid promising torque or efficiency results unless adequate measurement equipment is available.

1.3 Scope and Delimitations

Include:

- Define the first platform iteration and its electrical limits, including the intended DC-link voltage and current range.
- Identify the initial test motor and clarify that conclusions are limited to the investigated motor and operating range.
- List included work: requirements, hardware and PCB design, sensing, protection, firmware, modelling, simulation, control implementation, commissioning and experimental comparison.
- List excluded work: mechanical actuator development, gearbox design, flight qualification, production readiness, formal EMC and thermal qualification and high-voltage operation.
- Clarify that speed control is the principal common outer-loop objective. Torque and position control should be secondary or optional unless experimentally validated.

Writing guidance: Use one compact scope table with “included” and “excluded” columns.

1.4 Thesis Structure

Include:

- Provide one short paragraph describing the role of Chapters 2 to 9.

- Emphasize the logical sequence from theory and state of the art through methodology, modelling, implementation, verification, discussion and conclusion.
- Do not repeat the full table of contents.

Writing guidance: Keep this section to roughly half a page.

2 Theoretical Background

Chapter purpose: Present only the established theory needed to understand the model, platform design, control implementation and later interpretation. Project-specific solutions and measured results do not belong here.

2.1 Permanent-Magnet Synchronous Motors

Include:

- Introduce PMSM construction and the interaction between stator windings and permanent-magnet rotor.
- Explain mechanical angle, electrical angle, pole-pair count and their relationship.
- Describe back electromotive force, rotating magnetic fields and electromagnetic torque generation.
- Differentiate only the motor characteristics relevant to the chosen direct-drive high-pole-pair motor.
- Explain how motor parameters influence controller design and operating limits.

Writing guidance: Avoid an extensive taxonomy of electrical machines or manufacturing details not used later.

2.2 Mathematical Motor Model

Include:

- Present the phase-domain voltage equations and the mechanical dynamic equation.
- Introduce Clarke and Park transformations with the selected sign and axis conventions.
- Derive or state the d-q model used in simulation and FOC.
- Define flux linkage, phase resistance, d-axis and q-axis inductance, torque constant, inertia and friction where relevant.
- State assumptions such as sinusoidal back EMF, balanced phases and neglect of magnetic saturation if those assumptions are used.
- Connect each equation to its later use in modelling, controller design or result interpretation.

Writing guidance: Include only equations that are used later and define every symbol consistently in the nomenclature.

2.3 Three-Phase Inverter Operation

Include:

- Explain the three-phase bridge topology and half-bridge switching states.
- Describe complementary PWM, duty cycle, dead time and shoot-through prevention.
- Explain DC-link utilization and the relationship between switching commands and phase voltages.
- Introduce six-step switching, sinusoidal PWM and space-vector modulation only to the depth required by the actual implementations.
- Discuss switching frequency, switching losses and dead-time distortion as relevant implementation constraints.

Writing guidance: Use simplified topology and switching-state figures rather than complete project schematics.

2.4 Measurement and Rotor Position

Include:

- Explain in-line phase-current measurement using shunts and current-sense amplifiers.
- Describe phase-voltage acquisition and ADC scaling at a conceptual level.
- Explain magnetic rotor-position measurement, mechanical-to-electrical angle conversion, encoder direction and zero alignment.
- Describe offset, gain, bandwidth, quantization and sampling effects.
- Explain why synchronized current sampling is important for PWM-controlled drives.

Writing guidance: Keep circuit values and project-specific calibration equations for Chapters 6 and 7.

2.5 Motor Control Strategies

Include:

- Introduce a common classification of commutation and control methods.
- Define the inputs, outputs, sensing requirements and expected characteristics of the three selected methods.
- Provide a common conceptual basis for the detailed subsections below.

Writing guidance: Use a comparison table focused on principle, required feedback, parameter requirement and expected complexity.

2.5.1 Six-Step Commutation

Include:

- Explain electrical sectors, the commutation sequence and the active motor phases.
- Describe rotor-angle-based or Hall-like sector determination used by the planned implementation.
- Explain duty-cycle control, start-up behavior and current limitation.
- Discuss expected current waveform, torque ripple, acoustic behavior and low-speed limitations.
- State the main benefits: conceptual simplicity and low computational demand.

Writing guidance: A switching table or sector diagram is useful.

2.5.2 Sinusoidal Commutation

Include:

- Explain sinusoidal phase references generated from rotor electrical angle.
- Describe voltage-command or current-command variants and identify which one is relevant to the thesis.
- Explain PWM realization and the relationship between reference amplitude and torque production.
- Discuss smoother currents and torque compared with six-step control as well as the absence of full d-q current decoupling.
- Clarify the distinction from FOC to prevent conceptual overlap.

Writing guidance: Use one signal-flow diagram showing angle input, sinusoidal references and PWM generation.

2.5.3 Field-Oriented Control

Include:

- Explain d-axis and q-axis current components and their physical interpretation.
- Describe current measurement, Clarke transformation, Park transformation, current control, inverse transformation and modulation.
- Explain current references, voltage limitation, saturation and optional decoupling terms.
- Discuss required parameters, rotor-angle accuracy, tuning effort and computational demand.
- Relate FOC to expected low-speed behavior and current quality without presenting project results.

Writing guidance: Use a complete but readable FOC block diagram.

2.6 Cascaded Control

Include:

- Explain the cascade of fast inner control and slower outer speed control.
- Describe the speed controller and its output for each strategy, such as amplitude, torque-producing current or duty-cycle reference.
- Explain bandwidth separation and why the inner loop should respond faster than the speed loop.
- Introduce PI control, output limits and anti-windup.
- Define speed tracking measures used later, including rise time, overshoot, settling time and steady-state error.

Writing guidance: If torque and position loops are not evaluated, mention them briefly rather than creating separate theory sections.

2.7 Embedded Control Constraints

Include:

- Explain discrete sampling and digital controller execution.
- Describe PWM and ADC synchronization, interrupt timing and calculation delay.
- Discuss encoder update rates, speed estimation and numerical scaling.
- Introduce execution time, jitter, memory limits and missed-deadline risk.
- Explain why current-loop frequency and switching frequency constrain controller design.

Writing guidance: Link these constraints directly to later architectural and experimental decisions.

3 State of the Art

Chapter purpose: Critically review existing platforms, published comparisons and toolchains. The objective is to derive the specific gap addressed by the thesis rather than to provide a catalogue of products.

3.1 Existing Control Platforms

Include:

- Review representative commercial controllers, microcontroller evaluation boards and open research platforms.
- Compare openness, supported motors, sensing, configurability, accessible variables, protection functions and modification options.
- Distinguish industrial products from development boards and academic demonstrators.
- Identify limitations for connecting unfamiliar PMSMs and evaluating several algorithms on identical hardware.
- Include platforms only when they help justify a requirement or design decision.

Writing guidance: Use a criteria-based comparison table and cite authoritative documents or publications.

3.2 Comparative Control Studies

Include:

- Review studies comparing six-step, sinusoidal and field-oriented control or closely related methods.
- Extract the motors, hardware, operating conditions and metrics used in those studies.
- Assess whether comparisons used common hardware, common sampling conditions and fair tuning.
- Summarize established findings on speed tracking, low-speed behavior, current quality, torque ripple and computational effort.
- Identify inconsistencies or evidence gaps that motivate the experimental design of this thesis.

Writing guidance: Organize the section by comparison criteria rather than by author.

3.3 Development Tools and Frameworks

Include:

- Review relevant STM32 motor-control software, peripheral capabilities and available reference implementations.
- Discuss MATLAB and Simulink as tools for physical modelling, controller design and parameter tuning.
- Describe monitoring, communication and parameterization approaches relevant to laboratory commissioning.
- Clarify the role of vendor application notes, reference circuits and software libraries.
- Distinguish reusable external building blocks from the original contribution of the thesis.

Writing guidance: Do not treat vendor documentation as evidence for general scientific claims when peer-reviewed sources are available.

3.4 Research Gap and Contribution

Include:

- Synthesize the reviewed evidence into a concise research and engineering gap.
- Explain why existing platforms do not fully satisfy the need for transparent multi-strategy evaluation on common hardware.
- State the intended contribution: requirements-driven platform architecture, hardware and firmware integration, simulation-supported controller design and controlled experimental comparison.
- Explain what is not novel: the underlying control methods themselves.
- Connect the identified gap directly to the research objectives and methodology.

Writing guidance: This section should be an evidence-based synthesis, not a repetition of the introduction.

4 Methodology

Chapter purpose: Define the research logic before presenting implementation or results. Explain why the selected constructive and experimental methods are suitable for answering the research question.

4.1 Overall Research Approach

Include:

- Describe the combination of constructive engineering and controlled experimentation.
- Present the complete workflow from literature review and requirements through architecture, simulation, implementation, commissioning and comparison.
- Explain the distinction between verification of the platform and validation of its intended use.
- Define major decision gates and risk reductions, including simulation before hardware testing and use of a fallback inverter board.
- Show how evidence from each stage contributes to the final research answer.

Writing guidance: Include one methodology flowchart and a short explanation of each stage.

4.2 System Requirements

Include:

- Define intended users, use cases and system boundaries.
- Specify functional requirements for switching, sensing, rotor position, protection, control methods, speed control, monitoring and parameterization.
- Specify non-functional requirements for electrical ratings, modularity, real-time behavior, accuracy, reproducibility and extensibility.
- Assign identifiers and measurable acceptance criteria to principal requirements.
- Define the verification method for each requirement and place the full matrix in an appendix if necessary.

Writing guidance: Avoid vague requirements such as “fast” or “accurate”; use measurable criteria where possible.

4.3 Platform Development Method

Include:

- Explain how architecture alternatives and principal components are evaluated.
- Define the design sequence for schematic, review, PCB layout, manufacturing and assembly.
- Describe the firmware development sequence from peripheral drivers to common interfaces and control functions.
- Explain staged integration and risk management.
- Define documentation and version-control practices for hardware, firmware and parameter sets.
- Describe how prior project work is reused and attributed.

Writing guidance: Present the method, not the final hardware details.

4.4 Modelling and Simulation Method

Include:

- Define the purpose of simulation: model plausibility, controller design, tuning and reduction of commissioning risk.
- Specify model boundaries, input parameters, assumptions and excluded physical effects.
- Define solver type, sample times and the intended relationship between continuous plant model and discrete controller.
- Describe controller-design criteria and simulation test cases.
- Define how simulation outputs are assessed and which parameters are transferred to firmware.

Writing guidance: Avoid presenting simulation plots or tuned gains in this section.

4.5 Experimental Evaluation Method

Include:

- Define the independent variable as the control strategy.
- List controlled variables such as motor, inverter, DC-link voltage, current limit, speed profiles and data-processing method.
- Define dependent variables and metrics: tracking performance, current quality, low-speed behavior, execution time, memory use and tuning effort.
- Specify test cases, operating points, repetitions and data handling.
- Explain how fair tuning is achieved and documented.
- Define uncertainty, repeatability and validity considerations before running the tests.

Writing guidance: Use a test matrix that maps each test case to measured variables and comparison metrics.

5 Modelling and Simulation

Chapter purpose: Document the implemented model, controller design and supporting simulation evidence. Keep the focus on preparing and understanding the embedded implementation.

5.1 Motor Model Development

Include:

- Describe the implemented electrical and mechanical plant model.
- Document motor parameters, their sources and any identification procedures.
- Explain the inverter, load, friction and measurement representations included in the model.
- State parameter uncertainties and modelling assumptions.
- Demonstrate basic plausibility through steady-state or analytical checks.

Writing guidance: Separate inherited parameter data from parameters identified within the thesis.

5.2 Simulation Environment

Include:

- Describe the MATLAB and Simulink version, model organization and subsystem interfaces.
- Specify solver, step size, discrete sample times and numerical settings.
- Explain logging, parameter management and repeatable simulation execution.
- Describe saturation, voltage limits, current limits and fault logic included in the model.
- Show a top-level model diagram with readable interfaces.

Writing guidance: Avoid screenshots with unreadable block names; redraw key structures where necessary.

5.3 Control Strategy Models

Include:

- Explain the common simulation interfaces and the parts shared by all strategies.
- Document strategy-specific switching or modulation generation.
- Ensure that speed references, limits and evaluation signals are comparable across models.
- Describe how switching detail is represented and justify simplifications.

Writing guidance: The three subsections should use parallel structures for ease of comparison.

5.3.1 Six-Step Control Model

Include:

- Document sector generation, commutation table, duty-cycle command and current limiting.
- Explain any start-up or rotor-alignment assumptions.
- Show relevant signals and timing used to verify correct commutation.

Writing guidance: Focus on model behavior needed before embedded implementation.

5.3.2 Sinusoidal Control Model

Include:

- Document electrical-angle processing, reference generation and PWM mapping.
- Explain amplitude or current command generation and limiting.
- Show how the model distinguishes sinusoidal commutation from FOC.

Writing guidance: Report any simplifications in current dynamics or inverter switching.

5.3.3 Field-Oriented Control Model

Include:

- Document current sampling, Clarke and Park transforms, d-q controllers, voltage limits and inverse transforms.
- State reference values such as the d-axis current objective.
- Explain modulation and optional decoupling terms.

Writing guidance: Use consistent signs and units across equations, model and firmware.

5.4 Controller Design and Tuning

Include:

- Describe current-loop and speed-loop design methods.
- State bandwidth targets and explain their relation to switching and sampling frequency.
- Calculate or justify PI gains and anti-windup settings.
- Document saturation values, current limits and reference ramps.
- Explain tuning refinements made after simulation checks.
- Provide a final controller-parameter table for transfer to firmware.

Writing guidance: Separate analytically calculated initial parameters from empirically adjusted values.

5.5 Simulation Cases and Results

Include:

- Present a limited set of simulations that directly support implementation.
- Include model plausibility, current-loop response, speed-step response, current limitation and selected parameter sensitivities.
- Use identical axes and reference profiles when comparing strategies.
- Interpret the results in terms of controller readiness and expected commissioning behavior.
- Avoid treating simulation as a substitute for the final hardware comparison.

Writing guidance: End with a concise list of accepted parameters and unresolved model risks.

5.6 Transfer to Embedded Control

Include:

- Explain conversion of gains and limits into discrete embedded form.
- Document sample-time changes, scaling, units and numerical representation.
- Describe parameter storage and configuration interfaces.
- Identify differences between model and firmware, including dead time, quantization, measurement filtering and computational delay.
- Define checks used to ensure that transferred parameters are applied correctly.

Writing guidance: A model-to-firmware parameter mapping table is recommended.

6 Control Platform Design

Chapter purpose: Present the final architecture and its realization. Emphasize justified decisions and interfaces rather than a chronological account of development.

6.1 System Architecture

Include:

- Present the top-level hardware and software architecture.
- Show power flow, signal flow and protection paths.
- Define interfaces among host computer, STM32 controller, inverter, sensors, encoder and motor.
- Explain common and strategy-specific platform functions.
- Relate the architecture to the principal requirements.

Writing guidance: Use one overall architecture diagram as the reference for the remainder of the chapter.

6.2 Hardware Design

Include:

- Describe the custom three-phase inverter and principal component choices.
- Present the DC-link, gate-drive and power-stage design rationale.
- Document current and voltage measurement ranges, gains, filters and ADC scaling.
- Explain encoder interfacing and hardware protection paths.
- Describe PCB stack-up, grounding, power routing, signal partitioning, test points and connector concept.
- Include relevant calculations and selected schematic extracts while moving complete schematics to the appendix.

Writing guidance: For every major circuit, state requirement, alternatives, design decision, calculation and expected limitation.

6.3 Firmware Architecture

Include:

- Describe the software layers and module responsibilities.
- Present the real-time execution sequence, interrupt structure and loop rates.
- Document PWM generation, ADC triggering, encoder acquisition and speed calculation.
- Explain the system state machine, calibration, run, stop and fault states.
- Describe monitoring, communication and parameterization interfaces.
- Explain fault handling and safe-state behavior.

Writing guidance: Include a layered architecture diagram, state machine and timing diagram.

6.4 Control Algorithm Implementation

Include:

- Describe the common control interface and shared functions such as angle processing, speed calculation, current limiting, logging and fault handling.
- Explain how each strategy is selected and initialized.
- Document actual sample rates, numerical types and execution order.
- Identify use of external libraries or adapted reference code.

Writing guidance: Do not include long source-code listings in the main text.

6.4.1 Six-Step Control

Include:

- Document sector determination, switching table and duty-cycle generation.
- Explain commutation timing, direction reversal, start-up and current limitation.
- State implementation-specific constraints and protective measures.

Writing guidance: Include compact pseudocode or a flowchart only if it improves understanding.

6.4.2 Sinusoidal Control

Include:

- Document electrical-angle generation, sinusoidal references and PWM conversion.
- Explain amplitude commands, speed-loop integration and limitation behavior.
- State sampling and update rates.

Writing guidance: Clarify whether control is voltage-based or current-based.

6.4.3 Field-Oriented Control

Include:

- Document offset-corrected current acquisition, transformations, current controllers, voltage limitation and modulation.
- Explain rotor-angle alignment, d-q references and current-controller execution.
- State any omitted decoupling or compensation terms and justify them.

Writing guidance: Keep implementation notation consistent with Chapter 2 and Chapter 5.

6.4.4 Speed Control and Limitation

Include:

- Describe the common speed controller and strategy-specific output mappings.
- Document reference filtering or ramping, output saturation, current limits and anti-windup.
- Explain how comparable outer-loop behavior is maintained across strategies.
- State tuning differences that are necessary and how they are recorded.

Writing guidance: This section is central to the fairness of the later comparison.

6.5 System Integration

Include:

- Describe integration of hardware, firmware, control algorithms and host tools.
- Explain parameter transfer from simulation and final configuration management.
- Document resolved interface issues that materially changed the final design.
- Present the final platform configuration used for experiments.
- Distinguish planned functions from implemented functions.

Writing guidance: Avoid a project diary; include only iterations that explain the final system or a limitation.

6.6 Requirements Traceability

Include:

- Map principal requirements to hardware circuits, firmware modules and verification tests.
- State implementation status before testing: implemented, partial, deferred or not implemented.
- Identify deviations from initial requirements and justify accepted changes.
- Reference detailed evidence in Chapter 7 and the appendix.

Writing guidance: A concise traceability table is preferable to repetitive prose.

7 Experimental Verification and Evaluation

Chapter purpose: Provide objective evidence that the platform operates correctly and compare the three control methods under controlled conditions.

7.1 Platform Commissioning

Include:

- Present the safe staged commissioning sequence from unpowered checks to closed-loop motor operation.
- Document supply-rail checks, gate signals, complementary PWM, dead time and phase outputs.
- Explain current-limited initial operation and the criteria for progressing to each stage.
- Report major faults or design corrections only when they affect the final system or validity.
- Reference oscilloscope captures and test records.

Writing guidance: Correct title spelling: “Commissioning”. This section should not read as a daily laboratory log.

7.2 Sensor Calibration

Include:

- Describe current-sensor offset and gain calibration.
- Describe voltage-channel scaling and verification.
- Document encoder direction, zero alignment and electrical-angle offset.
- Verify speed calculation against an independent reference where possible.
- Quantify calibration residuals, repeatability and relevant uncertainty.
- State calibration conditions and whether recalibration is needed between runs.

Writing guidance: Provide equations, calibration plots and a concise results table.

7.3 Functional Platform Verification

Include:

- Verify PWM and phase-voltage generation.
- Verify synchronized current acquisition and plausible phase-current relationships.
- Verify rotor-position and speed acquisition.
- Measure loop frequency, execution time, jitter and deadline compliance.
- Test state transitions, parameter updates, monitoring and fault shutdown.
- Summarize requirement status as passed, partially passed, failed or not tested.

Writing guidance: Correct title spelling: “Platform”. Separate functional verification from comparative performance testing.

7.4 Experimental Setup

Include:

- Document the test motor, supply, inverter, encoder, measurement equipment and data-acquisition path.
- State equipment models, relevant accuracy, software versions and parameter-set versions.
- Describe mechanical mounting, environmental conditions and operating restrictions.
- Define test profiles, initial conditions, current limits, switching frequencies and sampling conditions.
- Explain repetitions and data-processing steps.
- Include a setup photograph or diagram and a complete test matrix.

Writing guidance: The description must enable a technically competent reader to reproduce the experiments.

7.5 Control Strategy Results

Include:

- Present results for each strategy using an identical structure and comparable plots.
- Separate direct observations from cross-strategy interpretation.
- Report unsuccessful operating points transparently.
- Use consistent time windows, axis ranges, units and filtering.

Writing guidance: Recommended sequence: start-up, steady state, low speed, dynamic response, current waveform, operating range and resource use.

7.5.1 Six-Step Control

Include:

- Present stable operation and sector commutation evidence.
- Report speed tracking and low-speed behavior.
- Show phase-current and phase-voltage waveforms.
- Report execution time, memory use and observed limitations.

Writing guidance: Discuss strategy-specific anomalies briefly; reserve general interpretation for Chapters 7.6 and 8.

7.5.2 Sinusoidal Control

Include:

- Present the same tests and plot formats used for six-step control.
- Show sinusoidal reference behavior and measured current quality.
- Report tracking performance, operating range and resource use.
- Document any angle or modulation limitations.

Writing guidance: Maintain consistent metrics to preserve comparability.

7.5.3 Field-Oriented Control

Include:

- Present current-loop evidence and d-q quantities where useful.
- Report speed tracking, low-speed behavior and phase-current quality.
- Show saturation or current-limit behavior when relevant.
- Report execution time, memory use and parameter sensitivity observed during testing.

Writing guidance: Avoid overwhelming the section with internal variables; select evidence that answers the research question.

7.6 Comparative Performance Assessment

Include:

- Compare rise time, overshoot, settling time and steady-state speed error.
- Compare speed ripple and minimum stable speed.
- Compare RMS current, peak current and ripple or THD if measurement quality permits.
- Compare worst-case execution time, memory use and missed deadlines.
- Assess required motor parameters, tuning effort and commissioning effort using transparent criteria.
- Present normalized plots and a weighted or unweighted decision matrix only if the scoring method is justified.
- Conclude which strategy is best suited to which evaluation task rather than declaring one universal winner.

Writing guidance: Use the same source data and processing method for all strategies.

7.7 Experimental Validity

Include:

- Assess measurement uncertainty, sensor bandwidth and calibration errors.
- Discuss repeatability and variability between runs.
- Assess fairness of controller tuning and differences in switching or sampling conditions.
- Discuss the single-motor limitation and restricted operating range.
- Explain the consequences of lacking calibrated torque instrumentation or a dynamometer.

- Compare measured and simulated behavior only to the extent supported by data.

Writing guidance: Distinguish internal validity, measurement validity and transferability.

8 Discussion

Chapter purpose: Interpret the engineering and scientific meaning of the results. Do not repeat every measurement; explain causes, consequences and confidence.

8.1 Platform Performance

Include:

- Assess whether the developed platform meets its intended use and principal requirements.
- Discuss hardware performance, sensing quality, protection behavior and real-time firmware.
- Evaluate modularity, observability, parameterization and effort required to connect another PMSM.
- Identify the most consequential design strengths and weaknesses.

Writing guidance: Use evidence from the traceability and verification results.

8.2 Simulation Accuracy

Include:

- Compare simulated and measured trends and selected quantitative responses.
- Explain deviations using parameter uncertainty, inverter nonidealities, filtering, quantization, delay and mechanical effects.
- Assess whether the model was sufficiently accurate for controller design and risk reduction.
- Identify model extensions that would materially improve prediction.

Writing guidance: Avoid framing imperfect agreement as failure if the model fulfilled its stated purpose.

8.3 Control Strategy Performance

Include:

- Explain why the strategies differed in speed tracking, current quality, low-speed behavior and computational demand.
- Relate observations to the theoretical background and prior comparative studies.
- Discuss the trade-off between simplicity, parameter demand, tuning effort and performance.
- Answer the supporting research questions and state use-case-specific recommendations.

Writing guidance: This is the main scientific interpretation of the comparative experiment.

8.4 Limitations and Transferability

Include:

- Synthesize hardware, software, modelling and experimental limitations.
- Explain how limitations constrain individual claims.
- Assess transferability to motors with different power, pole-pair count, inductance, back EMF or sensor characteristics.
- Distinguish limitations that affect absolute values from those that affect relative strategy ranking.
- Identify conclusions that are robust and conclusions that remain tentative.

Writing guidance: Use precise language and avoid overgeneralization.

8.5 Original Contribution

Include:

- Separate established methods and external building blocks from original work.
- Identify adapted vendor circuits, STM32 libraries and prior internal project results.
- State original requirements, architecture decisions, PCB integration, firmware structure and control integration.
- State the original experimental method, measurements and interpretation.
- Explain how the developed platform and evidence address the research gap.

Writing guidance: Correct numbering from 8.1.1-8.1.5 to 8.1-8.5 unless an 8.1 parent section is added.

9 Conclusion and Summary

Chapter purpose: Close the argument by answering the research question, identifying future work and providing any required end summary without repeating the abstracts.

9.1 Conclusion

Include:

- Restate the engineering problem and objective in one concise paragraph.
- Summarize the developed platform and implemented control methods.
- Present the most important quantitative verification and comparison results.
- Answer the primary research question directly.
- State whether the engineering objective was achieved and under which limitations.
- Provide a concise final recommendation for rapid PMSM evaluation.

Writing guidance: Do not introduce new evidence or literature. Prefer quantitative statements over general claims.

9.2 Future Work

Include:

- Prioritize work that directly follows from identified limitations.
- Consider testing additional PMSMs and broader operating ranges.
- Include improved parameter identification, torque measurement or dynamometer testing.
- Consider sensorless control, automated commissioning, improved HMI and standalone controller hardware.
- Include thermal characterization and EMC investigation where relevant.
- Separate necessary next steps from optional long-term extensions.

Writing guidance: Link each proposed activity to a limitation or opportunity established in Chapter 8.

9.3 Summary

Include:

- Provide a short standalone summary only if required by the supervisor, template or submission convention.
- Cover problem, method, developed system, principal findings and significance in a compact form.
- Avoid duplicating the conclusion word for word.
- If German and English abstracts already fulfil the summary requirement, consider omitting this section after consultation with the supervisors.

Writing guidance: A separate end summary is normally redundant when bilingual abstracts and a full conclusion are already present.

Appendix Planning

- Full requirements and verification matrix
- Complete schematics and selected PCB production files
- Detailed component calculations and data-sheet extracts where permitted
- Pin assignments and peripheral configuration tables
- Complete controller parameter sets
- Additional simulation plots and sensitivity runs
- Raw or supplementary measurement series
- Uncertainty calculations and equipment specifications
- Extended oscilloscope captures
- Source-code excerpts only where required for comprehension
- Documentation of reused prior work and third-party software components
- AI acknowledgement and tool-use documentation in the form required by the university

Principle: Keep evidence necessary to understand and evaluate the argument in the main text. Move large supporting datasets and implementation detail to appendices.